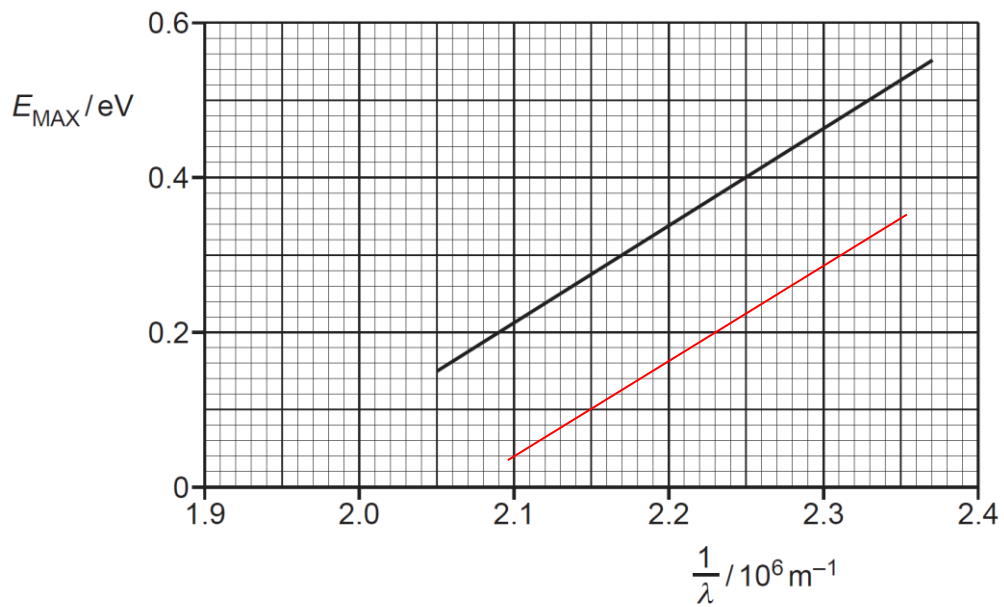


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| 6 | (a) | <p>When ultraviolet radiation of a specific frequency is incident on a metal surface, electrons are emitted with a range of kinetic energies up to a maximum value.</p> <p>Explain why the emitted electrons have a range of kinetic energies up to a maximum value.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     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|  |  |      |                                                                                                                                                                                                                                                                                                                                                             |                               |
|--|--|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
|  |  |      |                                                                                                                                                                                                                                                                                                                                                             |                               |
|  |  | (i)  | determine the threshold frequency $f_0$ of the metal.                                                                                                                                                                                                                                                                                                       |                               |
|  |  |      | $f_0 = \dots\dots\dots$ Hz                                                                                                                                                                                                                                                                                                                                  | [2]                           |
|  |  |      | <p><b>Solution:</b></p> <p>Extrapolate graph and find the x-intercept</p> <p>For <math>E_{\text{MAX}} = 0</math>, <math>\frac{1}{\lambda} = 1.93 \times 10^6</math></p> <p><math>f_0 = \frac{c}{\lambda_0} = 3.00 \times 10^8 \times 1.93 \times 10^6</math><br/> <math>= 5.79 \times 10^{14}</math> Hz</p>                                                 | <p>M1</p> <p>A1</p>           |
|  |  | (ii) | determine a value for the Planck constant $h$ . Explain your working clearly.                                                                                                                                                                                                                                                                               |                               |
|  |  |      | $h = \dots\dots\dots$ J s                                                                                                                                                                                                                                                                                                                                   | [3]                           |
|  |  |      | <p><b>Solution:</b></p> <p><math>E_{\text{MAX}} = \frac{hc}{\lambda} - \phi</math></p> <p>gradient = <math>hc</math></p> <p><math>= \frac{(0.4 - 0.2) \times 1.6 \times 10^{-19}}{(2.25 - 2.09) \times 10^6} = 2.0 \times 10^{-25}</math></p> <p><math>h = \frac{2.0 \times 10^{-25}}{3 \times 10^8}</math><br/> <math>= 6.7 \times 10^{-34}</math> J s</p> | <p>C1</p> <p>M1</p> <p>A1</p> |
|  |  | (c)  | <p>The electromagnetic radiation is now incident on a metal with a larger work function energy than the metal in (b).</p> <p>On Fig 6.1, sketch the variation with <math>\frac{1}{\lambda}</math> of <math>E_{\text{MAX}}</math>.</p>                                                                                                                       | [1]                           |

Solution:



With a larger work function, the threshold frequency required is larger and hence as  $f_0$  is proportional to  $\frac{1}{\lambda_0}$ , the value of the x axis-intercept is larger. The gradient  $hc$ , which is a constant, remains the same.

Straight line with same gradient as original and x-axis intercept greater than  $1.93 \times 10^6 \text{ m}^{-1}$ .

B1

(d) Infrared radiation of the same intensity is now incident on the same metal surface used in (b).

Explain why no electrons are emitted from the metal surface.

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.....[2]

[Total: 11]

Solution:

Infrared photons have **lower energy** than ultraviolet photons because they have a **lower frequency** since ( $E = hf$ ).

B1

The energy of an infrared photon is **less than the metal's work function**, so no electrons have enough energy to be emitted.

B1

